**MINOR PROJECT**

# Break-Even Analysis & Profitability Planning

Case study: a small cafe

**Submitted by:** Himanshu Kumar

**Subject area:** Finance (Cost-Volume-Profit Analysis)

**Date of submission:** 14 October 2026

# Contents

1. Project overview
2. Tools and technologies used
3. About the business and assumptions
4. Fixed and variable costs
5. Break-even point calculation
6. Impact of pricing changes
7. What-if analysis
8. Recommendations for profitability
9. Conclusion
10. Work evidence (screenshots)
11. External link and references

# 1. Project Overview

**Problem.** Many small businesses do not know how many units they must sell before they start making a profit. They set prices by guesswork or by copying competitors, and then find out after a few months that sales are not enough to cover their costs.

**Objective.** The aim of this project is to do a break-even analysis for a business and use it to support pricing and sales decisions. I have also used cost-volume-profit (CVP) analysis to see how a change in price, cost or sales volume changes the profit.

**What I did in this project:**

- Listed the fixed and variable costs of the business.
- Calculated the break-even point in orders and in rupees, along with the margin of safety.
- Tested what happens to break-even and profit when the selling price is changed.
- Checked how profit reacts when costs or sales change (what-if analysis).
- Gave suggestions to improve profitability.

### Basic formulas used

- Contribution per unit = Selling price per unit - Variable cost per unit
- Break-even point (units) = Total fixed cost / Contribution per unit
- Break-even sales (Rs) = Break-even units x Selling price
- Margin of safety (%) = (Expected sales - Break-even sales) / Expected sales x 100
- Profit = (Units sold x Contribution per unit) - Total fixed cost

# 2. Tools and Technologies Used

| Tool | Use in this project |
|---|---|
| Microsoft Excel | Cost sheet, break-even formulas, price scenarios, what-if table and the CVP data table |
| Python (matplotlib) | Drawing the CVP chart and the comparison charts shown in this report |
| PDF report | Final report prepared for submission |

The Excel file (*Minor_Project_BreakEven_Model.xlsx*) has five sheets: Inputs, BreakEven, PriceScenarios, Sensitivity and CVP_Data. All results in the sheets are calculated with formulas, so if any input is changed the whole model updates automatically.

# 3. About the Business and Assumptions

For this project I took a small cafe in a tier-2 city of India selling coffee, tea, snacks and light meals. **The figures used are my own assumptions based on typical costs for a small cafe. They are not taken from any real cafe's books.** The same method can be used for any business by replacing the inputs.

| Assumption | Value |
|---|---|
| Average selling price per order | Rs 180 |
| Expected orders per month | 2,100 (about 70 per day) |
| Operating days per month | 30 |
| Price elasticity of demand | -1.2 (assumed) |
| Target monthly profit | Rs 1,00,000 |

An elasticity of -1.2 means that if the price goes up by 10%, the number of orders is expected to fall by about 12%. This is only an assumption, because the real value depends on the cafe's customers and competition.

# 4. Fixed and Variable Costs

**Fixed costs** stay the same every month whether the cafe sells 500 orders or 3,000 orders. **Variable costs** change with the number of orders.

## Fixed costs (per month)

| Item | Amount (Rs) |
|---|---|
| Shop rent | 45,000 |
| Salaries (3 fixed staff) | 60,000 |
| Electricity (base charge) | 8,000 |
| Internet and phone | 1,500 |
| Licence and insurance | 2,500 |
| Equipment loan EMI | 15,000 |
| Marketing and social media | 6,000 |
| Repairs and miscellaneous | 4,000 |
| **Total fixed cost** | **1,42,000** |

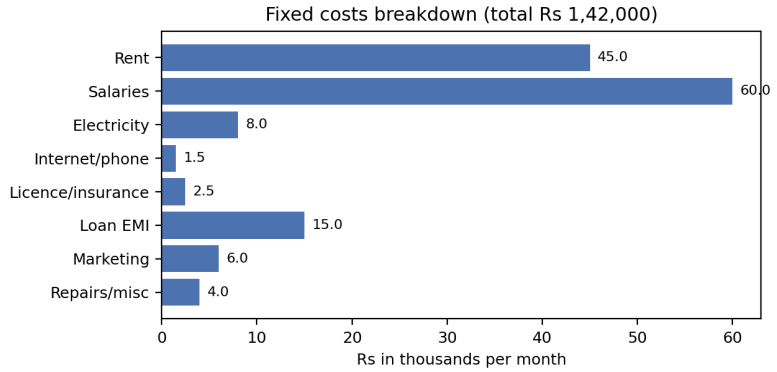



*Figure 1: Fixed costs breakdown*

Rent and salaries together are about 74% of the fixed cost, so these two items decide how risky the business is.

## Variable costs (per order)

| Item | Amount per order (Rs) |
|---|---|
| Ingredients (coffee, milk, food items) | 58 |
| Packaging | 7 |
| UPI / card charges | 3 |
| Gas and variable electricity | 4 |
| Wastage | 5 |
| **Total variable cost per order** | **77** |

With a selling price of Rs 180 and a variable cost of Rs 77, every order gives a contribution of Rs 103 towards fixed costs and profit.

## 5. Break-Even Point Calculation

**Contribution per order** = 180 - 77 = Rs 103
**Contribution margin ratio** = 103 / 180 = 57.2%
**Break-even point** = 1,42,000 / 103 = 1,378.6, rounded up to **1,379 orders per month**
**Break-even sales** = 1,379 x 180 = **Rs 2,48,220 per month**

| Measure | Result |
|---|---|
| Break-even orders per month | 1,379 |
| Break-even orders per day | about 46 |
| Break-even sales | Rs 2,48,220 |
| Expected sales (2,100 orders) | Rs 3,78,000 |
| Expected monthly profit | Rs 74,300 |
| Margin of safety | 721 orders (34.3%) |
| Orders needed for Rs 1,00,000 profit | 2,350 orders (about 78 per day) |
| Degree of operating leverage | 2.91 times |

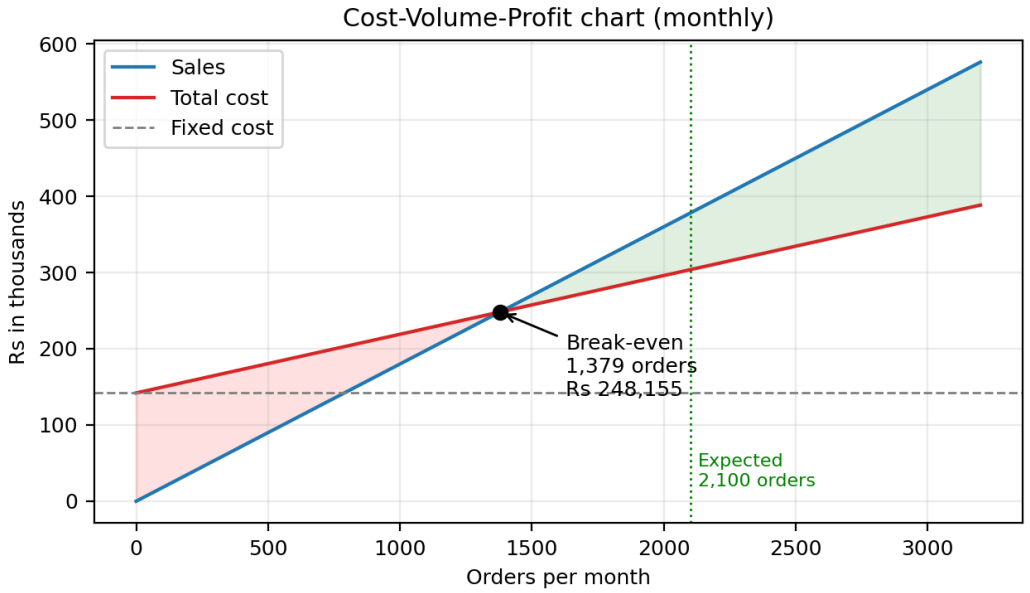



*Figure 2: Cost-Volume-Profit chart*

**What the results show.** The cafe needs about 46 orders a day just to cover its costs, and it expects around 70. This gives a margin of safety of 34.3%, which means sales can fall by about one third before the cafe starts making a loss. The operating leverage of 2.91 means that a 10% rise in sales would raise profit by roughly 29%, but a 10% fall in sales would also cut profit by the same amount. This is why fixed costs make the business risky.

## 6. Impact of Pricing Changes

I changed the price from -10% to +25% and checked the break-even point and profit. The table shows two views: (a) profit if the number of orders stays the same, and (b) profit if orders change as per the assumed elasticity of -1.2. Case (b) is more realistic.

| Price change | New price (Rs) | Contribution (Rs) | Break-even orders | Orders (with demand fall) | Profit (b) (Rs) | Profit (a) (Rs) |
|---|---|---|---|---|---|---|
| -10% | 162 | 85 | 1,671 | 2,352 | 57,920 | 36,500 |
| -5% | 171 | 94 | 1,511 | 2,226 | 67,244 | 55,400 |
| 0% | 180 | 103 | 1,379 | 2,100 | 74,300 | 74,300 |
| +5% | 189 | 112 | 1,268 | 1,974 | 79,088 | 93,200 |
| +10% | 198 | 121 | 1,174 | 1,848 | 81,608 | 112,100 |
| +15% | 207 | 130 | 1,093 | 1,722 | 81,860 | 131,000 |
| +20% | 216 | 139 | 1,022 | 1,596 | 79,844 | 149,900 |
| +25% | 225 | 148 | 960 | 1,470 | 75,560 | 168,800 |

*Table: price scenarios (base case shaded grey, best case shaded green)*

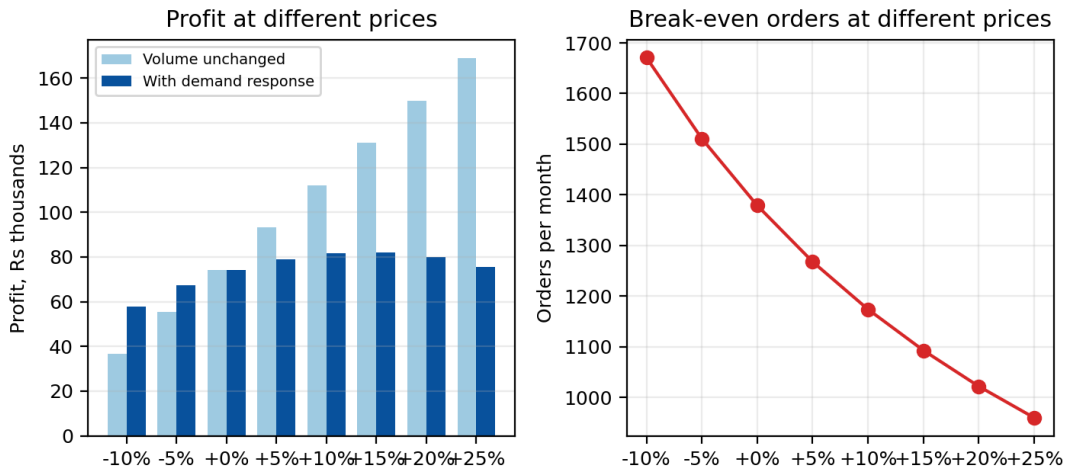



*Figure 3: Profit and break-even orders at different prices*

### Observations

- A 10% price cut raises break-even from 1,379 to 1,671 orders and lowers profit from Rs 74,300 to Rs 57,920, even though orders increase. The extra orders do not make up for the lower contribution per order.
- A 10% price rise lowers break-even to 1,174 orders and increases profit to about Rs 81,600 even after losing 12% of orders.
- In this model profit is highest at around +15% (Rs 207), at about Rs 81,860. Beyond that, the loss of customers is bigger than the gain from the higher price.
- Column (a) looks very attractive at high prices, but it is not realistic because customers will not stay the same if prices rise a lot.

- The result depends on the elasticity assumption. If customers are more sensitive to price (for example -2.0), a price rise will not be profitable. So the cafe should test a small price rise first.

## 7. What-If Analysis

I also checked how profit changes if costs or sales move by 10% to 20%.

| Scenario | Break-even orders | Monthly profit (Rs) |
|---|---|---|
| Base case | 1,379 | 74,300 |
| Variable cost up 10% | 1,491 | 58,130 |
| Variable cost down 10% | 1,283 | 90,470 |
| Fixed costs up 10% | 1,517 | 60,100 |
| Fixed costs down 10% | 1,241 | 88,500 |
| Orders fall by 20% | 1,379 | 31,040 |

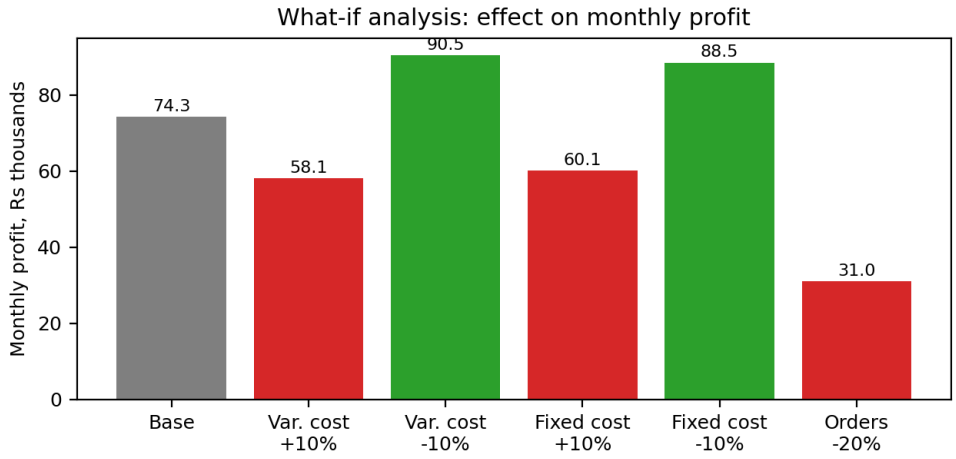



*Figure 4: Effect of cost and sales changes on profit*

A 20% fall in orders hurts the most. It reduces profit by more than half (from Rs 74,300 to Rs 31,040). A 10% rise in costs reduces profit by about Rs 14,000 to Rs 16,000. This shows that keeping customers coming is more important than small savings in cost.

# 8. Recommendations for Profitability

**1. Raise prices gradually.** A rise of 5% to 10% on popular items (for example Rs 189 to Rs 198 average) is likely to increase profit without losing too many customers. Customer reaction should be watched for 2 to 3 months before going further.

**2. Increase the average bill instead of only the price.** Combos such as coffee with a snack, or add-ons like extra shots, increase the amount spent per order. Because the variable cost of add-ons is low, most of the extra money becomes contribution.

**3. Reduce wastage and control ingredient cost.** Wastage and ingredients are Rs 63 of the Rs 77 variable cost. Buying in bulk from a steady supplier and tracking daily wastage can lower this. A 10% saving in variable cost adds about Rs 16,000 profit per month.

**4. Bring more customers on slow days and hours.** Since the cafe needs 46 orders a day to break even, off-peak offers, loyalty cards and tie-ups with nearby offices or colleges can raise volume at very little extra cost.

**5. Keep fixed costs under control.** Rent and salaries are 74% of fixed costs. Part-time staff in slow hours and negotiating a longer rent agreement can lower the fixed cost. Every Rs 10,000 cut in fixed cost reduces break-even by about 97 orders.

**6. Track the break-even point every month.** Costs change with time, so the break-even should be calculated again whenever rent, salary or ingredient prices change. The Excel model can be updated in a few minutes.

# 9. Conclusion

The break-even analysis shows that the cafe must sell about 1,379 orders (Rs 2,48,220 of sales) every month to cover its costs. At the expected 2,100 orders it earns around Rs 74,300 a month with a margin of safety of 34.3%. The price analysis shows that a small price rise gives higher profit than a price cut, as long as demand does not fall too sharply. Overall, I learned that profit depends on three things together: price, volume and cost structure, and that break-even analysis gives a clear number to plan around instead of guessing.

**Limitations.** The study uses assumed figures and a single average price per order. A real cafe has many items with different margins and its demand changes with season. The elasticity value is also an assumption. These can be improved with actual sales data.

# 10. Work Evidence (Screenshots)

The following screenshots are from the Excel workbook prepared for this project.

**Cafe Break-Even Analysis - Input Sheet**

*Blue cells are assumed inputs (typical small cafe in a tier-2 Indian city). Change them and every other sheet updates.*

| Fixed costs (per month) | Amount (Rs) |
|---|---|
| Shop rent | Rs 45,000 |
| Salaries (3 fixed staff) | Rs 60,000 |
| Electricity (base charge) | Rs 8,000 |
| Internet and phone | Rs 1,500 |
| Licence and insurance | Rs 2,500 |
| Equipment loan EMI | Rs 15,000 |
| Marketing and social media | Rs 6,000 |
| Repairs and miscellaneous | Rs 4,000 |
| **Total fixed cost** | **Rs 142,000** |

| Variable costs (per order) | Amount (Rs) |
|---|---|
| Ingredients (coffee, milk, food items) | Rs 58.00 |
| Packaging | Rs 7.00 |
| UPI / card charges | Rs 3.00 |
| Gas and variable electricity | Rs 4.00 |
| Wastage | Rs 5.00 |
| **Total variable cost per order** | **Rs 77.00** |

| Other assumptions | Value |
|---|---|
| Average selling price per order (Rs) | Rs 180.00 |
| Expected orders per month | 2,100 |
| Operating days per month | 30 |
| Price elasticity of demand (assumed) | -1.2 |
| Target monthly profit (Rs) | Rs 100,000 |

*Note: all figures above are assumptions made for this academic exercise, not data from a real cafe.*

*Screenshot 1: Inputs sheet (fixed costs, variable costs and assumptions)*

**Break-Even Calculation**

| Item | Value | Formula used |
|---|---|---|
| Total fixed cost per month | Rs 142,000 | *From Inputs* |
| Selling price per order | Rs 180.00 | *From Inputs* |
| Variable cost per order | Rs 77.00 | *From Inputs* |
| Contribution per order | Rs 103.00 | *Price - Variable cost* |
| Contribution margin ratio | 57.2% | *Contribution / Price* |
| **Break-even point (orders, exact)** | **1,378.6** | *Fixed cost / Contribution* |
| Break-even point (orders, rounded up) | 1,379 | *Round up to whole orders* |
| **Break-even sales (Rs)** | **Rs 248,220** | *BEP orders x Price* |
| Break-even orders per day | 46.0 | *BEP orders / Operating days* |
| Expected orders per month | 2,100 | *From Inputs* |
| Expected sales (Rs) | Rs 378,000 | *Orders x Price* |
| Total variable cost (Rs) | Rs 161,700 | *Orders x Variable cost* |
| **Expected monthly profit (Rs)** | **Rs 74,300** | *Sales - Variable - Fixed* |
| Margin of safety (orders) | 721 | *Expected - Break-even* |
| **Margin of safety (%)** | **34.3%** | *MoS orders / Expected orders* |
| Orders needed for target profit | 2,350 | *(Fixed + Target) / Contribution* |
| Degree of operating leverage | 2.91x | *Total contribution / Profit* |

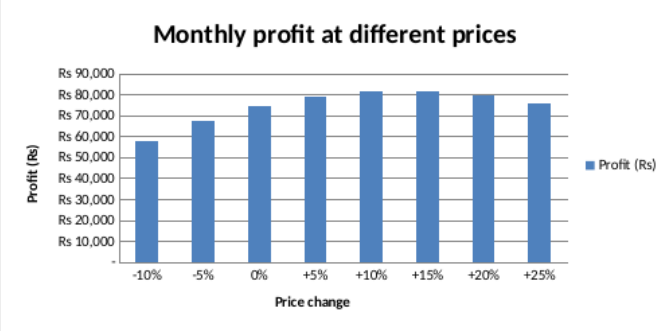
*Screenshot 2: BreakEven sheet (formula-based results)*

**Impact of Price Changes**

*Columns F-H assume demand reacts to price using the elasticity on the Inputs sheet. Column I assumes volume stays the same.*

| Price change | New price (Rs) | Contribution per order | Break-even orders | Break-even sales (Rs) | Expected orders (with demand response) | Sales (Rs) | Profit (Rs) | Profit if volume unchanged (Rs) |
|---|---|---|---|---|---|---|---|---|
| -10% | Rs 162.00 | Rs 85.00 | 1,671 | Rs 270,702 | 2,352 | Rs 381,024 | Rs 57,920 | Rs 36,500 |
| -5% | Rs 171.00 | Rs 94.00 | 1,511 | Rs 258,381 | 2,226 | Rs 380,646 | Rs 67,244 | Rs 55,400 |
| 0% | Rs 180.00 | Rs 103.00 | 1,379 | Rs 248,220 | 2,100 | Rs 378,000 | Rs 74,300 | Rs 74,300 |
| +5% | Rs 189.00 | Rs 112.00 | 1,268 | Rs 239,652 | 1,974 | Rs 373,086 | Rs 79,088 | Rs 93,200 |
| +10% | Rs 198.00 | Rs 121.00 | 1,174 | Rs 232,452 | 1,848 | Rs 365,904 | Rs 81,608 | Rs 112,100 |
| +15% | Rs 207.00 | Rs 130.00 | 1,093 | Rs 226,251 | 1,722 | Rs 356,454 | Rs 81,860 | Rs 131,000 |
| +20% | Rs 216.00 | Rs 139.00 | 1,022 | Rs 220,752 | 1,596 | Rs 344,736 | Rs 79,844 | Rs 149,900 |
| +25% | Rs 225.00 | Rs 148.00 | 960 | Rs 216,000 | 1,470 | Rs 330,750 | Rs 75,560 | Rs 168,800 |

**Best price (highest profit in table)** **Rs 207.00**
**Highest profit (Rs)** **Rs 81,860**




*Screenshot 3: PriceScenarios sheet*

# 11. External Link and References

**External link (GitHub / hosted):** Not applicable. The Excel file *Minor_Project_BreakEven_Model.xlsx* is submitted along with this report.

**References / concepts used:**

- Standard cost-volume-profit and break-even analysis from cost and management accounting textbooks.
- Common pricing strategy ideas (price-volume trade-off, bundling, upselling) from business case studies on small food and beverage businesses.
- All numerical values in this report are assumptions made by the student for demonstration.